

SECTION A

Q1 — 10. Wrong: (11) Si, Ge, Sn and Pb occur mainly combined, not free; (12) C and Si are not metals and Ge is a metalloid — none is a metal. The other ten are standard facts.

Q2 — 10. Wrong: (6) ΔH_1 does not fall smoothly — it rises slightly from Sn to Pb; (12) the electronegativities are not identical. Correct: large C→Si radius jump, small Si→Pb increase from poor d/f screening, $EN(Pb) > EN(Si)$ on the scale, C most electronegative.

Q3 — 10. Wrong: (7) the covalent radius of Ge (122 pm) is larger than Si (118 pm); (9) the elements do not all have identical electronegativity. Radii 77/118/122/140/146 pm, IE_1 1086/786/761/708/715 kJ mol⁻¹ (C > Si > Ge > Pb > Sn), diamond 3.51 / graphite 2.22, m.p. Sn < Pb, b.p. falling Si → Pb, $IE_1(Pb) > IE_1(Sn)$ all correct.

Q4 — 10. Wrong: (10) Pb(II), not Pb(IV), is the stable state; (11) the inert-pair effect becomes more important down the group.

Q5 — 10. Wrong: (10) the tetrachlorides are covalent; (12) PbI₄ does not exist. Correct: MO/MO₂, SiO only at high T, CO₂/SiO₂/GeO₂ acidic, SnO₂/PbO₂ and SnO/PbO amphoteric, CO neutral, C/Si/Ge inert to water, Sn + steam → SnO₂ + H₂, Pb oxide-film protected, MX₂ stability rising down the group.

Q6 — 10. Wrong: (7) heavier atoms form pπ–pπ bonds less effectively than carbon; (8) [SiCl₆]²⁻ is not formed — Cl⁻ is larger than F⁻ and too crowding.

Q7 — 10. Wrong: (5) catenation decreases down the group; (8) Pb shows essentially no catenation. Bond enthalpies 348/297/260/240 (so C–C > Si–Si and Sn–Sn < Ge–Ge) are correct.

Q8 — 10. Wrong: (6) diamond is an insulator — no delocalised electrons; (9) diamond is a 3-D network, not a layered van-der-Waals solid (that is graphite). “1 carat = 200 mg” is correct.

Q9 — 11. Wrong: (10) the layers are held by weak van der Waals forces, not covalent bonds.

Q10 — 12. Wrong: (10) C₆₀ does not contain only six-membered rings — 20 hexagons and 12 pentagons. “Both single and double C–C bonds” (143.5 / 138.3 pm) is correct.

Q11 — 11. Wrong: (9) 1 carat = 200 mg, not 20 mg.

Q12 — 10. Wrong: (7) water gas is CO + H₂; (9) excess-O₂ combustion gives CO₂ — pure CO comes from HCOOH + conc. H₂SO₄.

Q13 — 10. Wrong: (8) CO is a neutral oxide; (10) CO, not CO₂, forms carboxyhaemoglobin.

Q14 — 11. Wrong: (9) CO₂ is an acidic oxide. Blood pH 7.26–7.42 is correct.

Q15 — 11. Wrong: (10) CO₂ is linear (sp carbon, no lone pair), not bent.

Q16 — 11. Wrong: (12) SiO₂ does not dissolve in ordinary mineral acids — only HF (and hot NaOH) attack it.

Q17 — 10. Wrong: (9) R₄Si has no hydrolysable Si–Cl and is not a cross-linker (RSiCl₃ is); (10) the Si–O–Si backbone defines silicones.

Q18 — 10. Wrong: (10) silicones are hydrophobic (no free –OH); (11) they have high dielectric strength.

Q19 — 10. Wrong: (10) tetrahedra polymerise by corner-sharing, not edge-sharing; (11) silicates are not limited to isolated ions.

Q20 — 10. Wrong: (7) Al-for-Si substitution makes the framework negative; (8) softening is by cation exchange.

SECTION B

Q21 — C. II is wrong: silicon, not carbon, is the 2nd most abundant crustal element. I, III, IV, V are standard facts.

Q22 — A. V is wrong: ΔH_1 does not fall smoothly — Pb lies above Sn. I–IV describe the radius jump, poor d/f screening, the Sn minimum, and Group 14 > Group 13 in ΔH_1 .

Q23 — B. V is wrong: excess CO₂ dissolves the CaCO₃ precipitate (as Ca(HCO₃)₂), it does not fix it. I–IV are correctly balanced.

Q24 — B. II is wrong: Pb(II), not Pb(IV), is the stable state. I, III, IV, V (inert-pair trend, Sn(II) reducing / Pb(IV) oxidising, Pb⁴⁺ > Sn⁴⁺ as oxidant) are correct.

Q25 — D. All five are genuine consequences of the inert-pair effect (PbO₂ oxidiser, no Pb catenation, no PbI₄, GeO₂ acidic / GeO amphoteric, PbCl₄ → PbCl₂ + Cl₂).

Q26 — A. Only sets I (Cr₂O₃, BeO, SnO, SnO₂) and IV (ZnO, Al₂O₃, PbO, PbO₂) contain only amphoteric oxides. CrO and NO are not amphoteric; B₂O₃, SiO₂, CO₂ and GeO₂ are acidic.

Q27 — A. V is wrong: CO is a neutral oxide and does not give H₂CO₃. I–IV cover MO/MO₂ formation, acidic CO₂/SiO₂/GeO₂ with acidity CO₂ > SiO₂ > GeO₂, amphoteric SnO₂/PbO₂, neutral CO.

Q28 — A. V is wrong: CCl₄ resists hydrolysis. I–IV: C/Si/Ge inert, Sn + steam, Pb film; all MX₄ except CCl₄ hydrolyse; Si expands coordination; SnF₄/PbF₄ ionic.

Q29 — D. All correct: six-coordinate [SiF₆]²⁻/[GeCl₆]²⁻/[Sn(OH)₆]²⁻; [SiCl₆]²⁻ not formed; sp³d² description; no [CF₆]²⁻/[CCl₆]²⁻; SiF₄ + F⁻ → [SiF₆]²⁻ but CF₄ unreactive.

Q30 — B. III is wrong: catenation is far greater for carbon and falls sharply down the group. I, II, IV, V (anomaly causes, covalence 4, order C >> Si > Ge ≈ Sn, bond enthalpies) are correct.

Q31 — A. V is wrong: Si forms weaker pπ–pπ bonds to O than C. I–IV: planar N(SiH₃)₃, wide (SiH₃)₂O angle, dπ back-donation, trisilylamine a weaker base.

Q32 — A. III is wrong: a layered van-der-Waals solid is graphite. I, II, IV, V are the standard diamond facts.

Q33 — B. III is wrong: graphite is layered (not a 3-D network) and is not an abrasive. I, II, IV, V (interlayer 340 pm, 141.5 pm sp^2 , conduction, lubricant/electrodes) are correct.

Q34 — C. IV is wrong: a five-membered ring fuses only with six-membered rings (isolated-pentagon rule). I, II, III, V are correct (arc preparation, no dangling bonds, $20 + 12$ rings, sp^2 , $143.5/138.3$ pm).

Q35 — A. V is wrong: fullerenes have no dangling bonds. I–IV: non-planar cages, 6:6 bonds shorter, less aromatic than benzene, 12 pentagons with $(n - 20)/2$ hexagons.

Q36 — C. II is wrong: C_{60} has 12 pentagons and 20 hexagons, not the reverse. I, III, IV, V: ΔfH° values, distances, $\Delta H < 0 / \Delta S > 0$ for diamond $\rightarrow$ graphite, 32 rings + 90 edges.

Q37 — C. IV is wrong: a “lead” pencil is graphite + clay, with no metallic lead. I, II, III, V are correct.

Q38 — C. III is wrong: water gas is $CO + H_2$, not $CO_2 + H_2$. I, II, IV, V: incomplete-combustion/HCOOH route, water and producer gas, producer gas poorer, water-gas shift.

Q39 — A. V is wrong: CO is a neutral oxide. I–IV: isoelectronic with $N_2/CN^-/NO^+$, C-donor with π^* back-bonding, reduces Fe_2O_3/ZnO , $\sim 300\times$ stable carboxyhaemoglobin.

Q40 — A. V is wrong: carbonic acid is weak. I–IV: lab/commercial routes, weak dibasic H_2CO_3 , blood buffer 7.26–7.42, carbonate vs bicarbonate by ratio.

Q41 — B. III is wrong: CO_2 is linear. I, II, IV, V: sp carbon, 115 pm, zero dipole, isostructural with N_3^-/N_2O ; dry ice / fire extinguisher / urea; photosynthesis and greenhouse; molecular solid from good $\pi\pi$ - $\pi\pi$ bonding.

Q42 — A. V is wrong: red-hot carbon + steam gives water gas ($CO + H_2$), not $CO_2 + H_2$. I–IV are correctly balanced.

Q43 — A. V is wrong: formic acid + conc. H_2SO_4 gives CO. I–IV: C_3O_2 from malonic acid (4 σ + 4 π), phosgene $\rightarrow$ urea, oxalic acid $\rightarrow CO + CO_2 + H_2O$.

Q44 — C. III is wrong: WC is a hard refractory carbide, insoluble in water. I, II, IV, V distinguish acetylide/methanide/allylide carbides and network/interstitial types, plus the cyanamide reaction.

Q45 — A. Sn, Fe and Zn oxides are reduced by carbon/CO in metallurgy; Al_2O_3 and $MgCO_3 \cdot CaCO_3$ give metals too electropositive for carbon reduction — Al and Mg are obtained by electrolysis. (III and IV.)

Q46 — C. III is wrong: SiO_2 has no discrete $Si=O$ bonds — it is an $Si-O-Si$ network. I, II, IV, V: polymorphs and kieselguhr, tetrahedral network, $HF/NaOH$ attack only, piezoelectric quartz and silica gel.

Q47 — D. All five are correct: direct process at 573 K/Cu; $R_2SiCl_2 \rightarrow$ linear/cyclic; $RSiCl_3$ trifunctional cross-linker; R_3SiCl monofunctional end-blocker; R_4Si has no hydrolysable $Si-Cl$.

Q48 — B. III is wrong: silicones are hydrophobic with high dielectric strength. I, II, IV, V: non-polar shielding, thermal/oxidation resistance, sealants/greases/insulators/water-proofing/antifoaming, biomedical implants.

Q49 — A. IV is wrong: silicates range from isolated $[SiO_4]^{4-}$ to 3-D frameworks. I, II, III, V: tetrahedral unit, 1–4 corner sharing, cation charge balance and mineral examples, 4 shared corners $\rightarrow SiO_2$.

Q50 — C. V is wrong: a single chain has $O:Si = 3$ while a sheet has $O:Si = 2.5$. I–IV give the correct formula ladder ortho $\rightarrow$ pyro $\rightarrow$ ring $\rightarrow$ chain $\rightarrow$ double chain $\rightarrow$ sheet $\rightarrow$ framework.

Q51 — C. V is wrong: CCl_4 boils below $SiCl_4$, and graphite is softer than diamond. I–IV: catenation, IE_1 , oxide-acidity and hydride-reducing orders and MX_2 stability rising down the group.

Q52 — A. IV is wrong: the framework is negatively charged and softening is by cation exchange. I, II, III, V: negative aluminosilicate framework, ZSM-5 catalysis, cation-exchange softening, silicic-acid gel from $Na_2SiO_3 + HCl$.

Q53 — C. V is wrong: tin + conc. H_2SO_4 gives SO_2 , not H_2S . I–IV: tin pest, the oxide/salt names, lead ores and chrome yellow, Sn/Pb solder.

Q54 — C. V is wrong: toward conc. HCl , PbO_2 oxidises Cl^- to Cl_2 — it is the oxidant. I–IV: $Pb_3O_4 \equiv 2PbO \cdot PbO_2$ and its reactions, $Pb(NO_3)_2$ decomposition, $SnCl_2$ reducing $FeCl_3$ and $HgCl_2$.

SECTION C

Q55 — D. Catenation is greatest for carbon (strongest E–E bond, 348 kJ mol^{-1}) and falls sharply down the group.

Q56 — B. White (β) tin transforms to powdery grey (α) tin below $\sim 13^\circ C$ — an allotropic change, i.e. a change in the crystalline structure.

Q57 — C. CO , N_2 , CN^- and NO^+ are isoelectronic (14 e^-) with bond order 3. NO has bond order 2.5, O_2 has 2, O_2^- has 1.5.

Q58 — C. $SiCl_4 + 4H_2O \rightarrow H_4SiO_4 (X) + 4HCl$; $H_4SiO_4 -\Delta$, $\sim 1000 \text{ K} \rightarrow SiO_2 (Y) + 2H_2O$.

Q59 — B. In every single-chain (metasilicate) silicate — whatever its repeat period 1T, 2T, 3T or 4T — each SiO_4 tetrahedron shares two of its four oxygens with neighbours, so the repeat anion is $(SiO_3)_n^{2n-}$ ($O : Si = 3$). The “nT” label only fixes the geometric repeat period, not the connectivity or $O : Si$ ratio.

Q60 — A. B and C are second-period elements with no d orbitals in the valence shell, so they cannot expand their octet to accommodate six ligands; Al and Si (period 3) have accessible 3d orbitals.

Q61 — C. “Dry ice” is solid carbon dioxide, which sublimates at $-78^\circ C$.

Q62 — D. Only carbon, being small with compact 2p orbitals, forms effective $\pi\pi$ - $\pi\pi$ multiple bonds; the heavier congeners do so poorly.

Q63 — C. $Sn + 2H_2O(g) \xrightarrow{\text{hot}} SnO_2 + 2H_2$. Si and Ge are unaffected; Pb is protected by an oxide film.

Q64 — D. A is false — the heavier elements fail to form $\pi\pi$ - $\pi\pi$ bonds precisely because their orbitals are large and diffuse (poor overlap). B, C, D are true $\rightarrow F T T T$.

Q65 — C. CCl_4 has no low-energy route for attack at carbon (no d orbitals, no lone pair) and resists hydrolysis; SiCl_4 hydrolyses readily by expanding silicon's coordination number.

Q66 — D. Malonic acid ($\text{HOOC}-\text{CH}_2-\text{COOH}$) loses $2\text{H}_2\text{O}$ to give carbon suboxide, C_3O_2 ($\text{O}=\text{C}=\text{C}=\text{O}$).

Q67 — B. The difunctional Me_2SiCl_2 builds the siloxane chain; the monofunctional Me_3SiCl caps both ends $\rightarrow$ a short, end-blocked linear silicone. (MeSiCl_3 would instead give a cross-linked network.)

Q68 — D. Kaolinite is $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$ — a 1:1 layer of the $(\text{Si}_2\text{O}_5)_n^{2n-}$ sheet (3 shared O per tetrahedron) bonded to a gibbsite-type $\text{Al}(\text{OH})_3$ sheet.

Q69 — C. A finite set of tetrahedra each sharing exactly two corners closes into a ring; three units give $\text{Si}_3\text{O}_9^{6-}$ (3 bridging + 6 terminal O, each terminal O -1).

Q70 — D. Litharge is PbO (yellow); red lead is Pb_3O_4 ($= 2\text{PbO} \cdot \text{PbO}_2$, bright red).

SECTION D

Q71 — 13. Every fullerene has exactly 12 pentagons ($x = 12$); for C_{70} the hexagon count is $(70 - 20)/2 = 25$ ($y = 25$). $y - x = 13$.

Q72 — 65. Densities $\approx$ graphite 2.2, diamond 3.5 $\rightarrow (2.2 + 3.5) \cdot 10 = 57 = \text{A}$. Al_2Cl_6 has 2 Cl $\rightarrow$ Al dative bonds ($\text{P} = 2$); $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ has 6 O $\rightarrow$ Al dative bonds ($\text{Q} = 6$). $\text{A} + \text{P} + \text{Q} = 65$.

Q73 — 4. IE_1 : B 801, C 1086, Si 786, Ge 762 — all above Pb (715); Al, Ga, In, Tl and Sn are all below. Count = 4.

Q74 — 12. The six-membered ring is $[\text{Si}_6\text{O}_{18}]^{12-}$: $x = 6$, $y = 18$, $z = 12 \rightarrow |6 + 18 - 12| = 12$.

Q75 — 7. sp carbon in CO_2 , C_3O_2 , $\text{CO} \rightarrow x = 3$. Be_2C is a methanide ($p = 1$); Al_4C_3 is the aluminium methanide ($q = 3$). $x + p + q = 7$.

Q76 — 5. (i)–(v) are correctly matched; (vi) is wrong (PbO is amphoteric, not acidic). Count = 5.

Q77 — 6. Neutral CO ($x = 1$); basic none ($y = 0$); acidic SiO_2 , B_2O_3 , CO_2 ($z = 3$); amphoteric PbO_2 , SnO_2 , PbO , SnO , Al_2O_3 ($t = 5$). $|2 + 0 - 3 - 5| = 6$.

Q78 — 8. In $[\text{Al}(\text{BH}_4)_4]^-$ each BH_4 is bidentate, contributing two Al–H–B three-centre bonds; $4 \times 2 = 8$.

Q79 — 4. Colemanite ($\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O}$) has, per B_6 unit, four tetrahedral BO_4 boron atoms (no vacant p-orbital, hence no back bonding) and two trigonal BO_3 boron. Non-back-bonded = 4.

Q80 — 4. ≥ 2 shared O per tetrahedron in: beryl ring $\text{Be}_3\text{Al}_2[\text{Si}_6\text{O}_{18}]$ (2), pyroxene $\text{LiAlSi}_2\text{O}_6$ (2), amphibole Si_4O_{11} (avg 2.5), sheet Si_2O_5 (3). $\text{Sc}_2[\text{Si}_2\text{O}_7]$ shares only one. Count = 4.

Q81 — 7. Paramagnetic: Fe_2O_3 ($\text{Fe}^{3+} d^5$), MnO ($\text{Mn}^{2+} d^5$), NO (odd electron), PmO , Eu_2O_3 , U_2O_3 (unpaired f electrons), MnO_2 ($\text{Mn}^{4+} d^3$). Count = 7.

Q82 — 9. Every closed fullerene has exactly 12 pentagons $\rightarrow 12 + 12 + 12 = 36 \rightarrow$ digit sum $3 + 6 = 9$.

SECTION E

Q83 — C. Water gas contains $\text{CO} + \text{H}_2$ (P,R); producer gas contains $\text{CO} + \text{N}_2$ (P,Q); coal gas contains CO , H_2 and CH_4 (P,R,S); natural gas is mainly CH_4 (S).

Q84 — B. All are built from SiO_4 tetrahedra (P). Orthosilicate shares no corners (Q); pyrosilicate shares one corner per tetrahedron (R); a pyroxene single chain shares two corners and is polymeric (S,U); a mica sheet shares three corners and is polymeric (T,U).

Q85 — D. CO_2 has sp carbon and is linear (P,Q); CO_3^{2-} has sp^2 carbon and trigonal-planar geometry (R,S); $[\text{SiF}_6]^{2-}$ is six-coordinate and is described as sp^3d^2 (V,W); SiCl_4 is sp^3 and tetrahedral (T,U).

Q86 — A. CO is a neutral monoxide (P,T); CO_2 is acidic (Q); SnO is an amphoteric monoxide (R,T); PbO_2 is amphoteric and a strong oxidising agent (R,S).

Q87 — C. CCl_4 and SiCl_4 are tetrahedral molecular halides (P). CCl_4 resists hydrolysis (Q), whereas SiCl_4 hydrolyses readily (R). $[\text{SiF}_6]^{2-}$ and $[\text{GeCl}_6]^{2-}$ are six-coordinate species with coordination number greater than four (S,T).

Q88 — B. Diamond is a three-dimensional sp^3 covalent network (P,U). Graphite is sp^2 with delocalised electrons responsible for conduction (Q,R). C_{60} is an sp^2 cage molecule (Q,S). Carbon black is used as a pigment and tyre filler (T).

Q89 — D. CO is a neutral oxide and reducing agent (P,R). CO_2 is acidic (Q). H_2CO_3 is a weak dibasic acid (T). Water gas contains CO as the reducing component and H_2 as a major constituent (R,S).

Q90 — A. R_2SiCl_2 contains hydrolysable Si–Cl bonds and mainly gives linear/cyclic siloxanes (P,Q). RSiCl_3 is hydrolysable and introduces cross-linking (P,R). R_3SiCl is hydrolysable and can terminate chains (P,S). R_4Si has no hydrolysable Si–Cl bond (T).

Q91 — C. SnCl_2 acts as a reducing agent (P). PbO_2 is amphoteric and oxidising (Q,R). Pb_3O_4 is red lead and can be written as a mixed Pb(II)/Pb(IV) oxide (S,U). PbO is amphoteric and is litharge (R,T).

Q92 — B. $[\text{SiO}_4]^{4-}$ is an isolated tetrahedron (P); $[\text{Si}_2\text{O}_7]^{6-}$ has one shared corner per tetrahedron (Q); $[\text{Si}_3\text{O}_9]^{6-}$ is cyclic with two shared corners per tetrahedron (R,T); $(\text{Si}_2\text{O}_5)_n^{2n-}$ is a sheet with three shared corners per tetrahedron (S,U).

SECTION F

Q93 — A. The bright red solid is red lead, Pb_3O_4 . Warm dilute HNO_3 attacks only its “ PbO ” part: $\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \rightarrow 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 \downarrow + 2\text{H}_2\text{O}$, leaving dark brown PbO_2 as B.

Q94 — A. With conc. HCl the PbO_2 part oxidises chloride: $\text{Pb}_3\text{O}_4 + 8\text{HCl} \rightarrow 3\text{PbCl}_2 + \text{Cl}_2 \uparrow + 4\text{H}_2\text{O}$; the white residue C is PbCl_2 .

Q95 — C. A lone tetrahedron is orthosilicate SiO_4^{4-} ; three tetrahedra in a closed ring, each sharing 2 corners, give $\text{Si}_3\text{O}_9^{6-}$ (3 bridging + 6 terminal O).

Q96 — C. A trisilicate (Si_3O_{10}) chain has charge 8^- ; with Ca^{2+} : Cu^{2+} : $\text{H}_2\text{O} = 2 : 2 : 2$ the neutral formula is $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$ — the mineral kinoite.